Taxes, Spending, and Property Values: Supply Adjustment in a Tiebout-Oates Model

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This paper extends the capitalization approach of Wallace E. Oates to consider supply adjustment in a local public goods market of the sort hypothesized by Charles M. Tiebout. It is shown that Oates's single-period cross-section analysis-demonstrated demand conditions approximated Tiebout's hypotheses in a situation in which supply was not in long-run equilibrium. Analysis of capitalization of taxes, school expenditures, and road maintenance expenditures over five successive census periods in the Boston area indicates a move toward equilibrium in the market for schooling, but not in all other markets for local public goods.

I

Nearly 2 decades ago, Charles M. Tiebout (1956) proposed his well-known hypothesis that provision of local government services "reflects the preferences of the population more adequately than they can be reflected at the national level." In response to the arguments of Samuelson (1954) and Musgrave (1959) that no market test existed for the optimal level of (national) public goods output, Tiebout proposed a model under which a market analogue could lead to optimal expenditure on local public goods. The Tiebout model has since become a source of support for local

A preliminary version of this paper was presented at the annual meetings of the Western Economic Association, August 16, 1973. Research was funded by the Economics Division of the National Science Foundation and by the Center for the Study of Metropolitan Problems (Metro Center), National Institute of Mental Health. The comments of Ronald Grieson and the assistance of Eugene Kroch and Herman Thomas are acknowledged with thanks.
[*Journal of Political Economy*, 1974, vol. 82, no. 5]

fiscal autonomy, and is frequently cited in discussions of revenue sharing and in opposition to metropolitan consolidations. This has been particularly the case since Wallace E. Oates's (1969) publication of an "empirical study of tax capitalization and the Tiebout hypothesis," which is often interpreted as confirming the Tiebout model.

This paper extends Oates's capitalization approach to consider supply adjustment in the local public goods market. We argue that Oates restated Tiebout's model into a simpler question of whether households have preferences for a tax-service mix. This formulation does not consider whether the supply of public service combinations approaches the elasticity required for the model to yield Tiebout's analogue to the perfectly competitive solution. Oates verified that consumer demand conforms to Tiebout's hypothesis. However, this result was found only because supply was not in long-run equilibrium. If supply conditions in all markets had attained the competitive equilibrium, no capitalization would have been measured.

To investigate further the dynamics of an Oates-type market, we relate house values to local taxes and service delivery over different time periods for the Boston area. For schooling, there was a movement toward competitive equilibrium between 1950 and 1970. However, both continued tax capitalization despite the declining capitalization of school-expenditure differentials, and the continuing failure of highway-expenditure differentials to be capitalized indicate that supply conditions are not moving toward equilibrium in the markets for all public services. These empirical findings indicate that the Tiebout hypothesis cannot be considered as verified for local public service markets in general but only for some particular local service markets.

II

We begin by considering the manner in which local service benefits might be capitalized into land values, or location rents for sites in different tax-service districts, in a system characterized by the Tiebout model. In particular, we may assume that Tiebout's first five assumptions either are met directly in the system we are considering, or affect land values in a purely additive manner if they are unmet. That is, we assume (1) mobility and (2) knowledge among a population of utility-maximizing residents, (3) sufficient numbers of communities to insure competition among them and the availability of communities for each consumer's taste, (4) absence of restrictions on location due to employment needs (or, at least, differentiation among communities on grounds of accessibility to jobs that will enter into land values as a simple function of distance), and (5) absence of externalities between communities from public services.

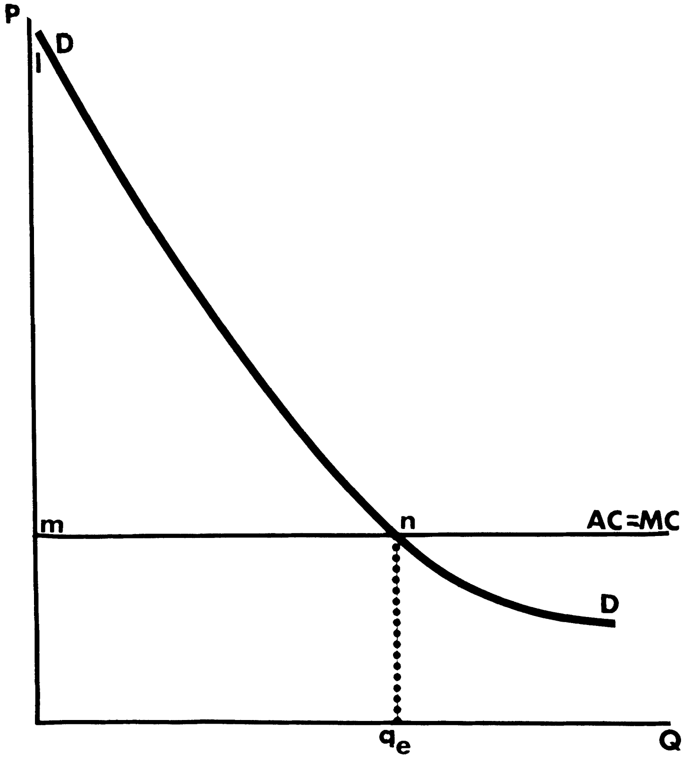

FIG. 1.—Supply and demand in a market for secular junior high school educations

Since we are assuming an analogue to a perfect market, point 3 implies that the residents of each town will be homogeneous with respect to their taste for each public service. Thus, for example, if education is provided, families without children, those with few children, and those with many children will be able to find towns fitting their needs and identical in all respects except for the provision of schooling. In addition, we begin with the assumption—which Tiebout approximates in the sixth and seventh conditions[1]—that constant returns to scale prevail in the provision of the service, or that, at least, equilibrium is possible with each town at a minimum-cost scale. For the empirical results presented later to hold it is also necessary to assume that funding is based on a local property tax rather than state or federal subsidies, which would add to benefits in some towns in a nonrandom manner.

In this situation, how will a market be represented in town provision of services? In figure 1, the cost of providing some public service (e.g.,

[1] The actual assumptions used by Tiebout are (6) that for each pattern of consumer wants there is an optimal community size (which yields least costs), and (7) that communities try to attain the optimal size to minimize average costs.

secular junior high school educations) is represented as \$$m$/unit (where a unit is, say, a student educated); DD is the demand curve for these junior high school educations. Since the average cost is passed on to consumers as a local tax in towns with these schools, it appears to them as a price. As many people move to towns with schools as are willing to pay a tax of m per average number of children per family. Since average cost, tax per pupil, and marginal cost are all the same, the market result is efficient in that marginal demand (marginal utility) is equated to marginal cost. As long as either there are true constant returns to scale, or the quantity demanded q_e is multiple of the optimal size, minimum cost is also insured.

In this case, the consumers of secular public junior high schools will reap a total consumer surplus of *lmn*. The schools will cover their costs. But will there be a corresponding producers' surplus to be capitalized into land values? Assuming a sufficient number of towns that anyone can find an otherwise acceptable location either with or without a secular junior high school, no resident would offer an after-tax rent above that attributable to nonschool-oriented attributes of the town. Apart from the additional tax paid for the schools, no resident would pay more than anyone locating in a school-less town to avoid the tax because of a lack of children or a preference for religious junior high schools (which may collect their average costs under the name of tuition fees rather than taxes). As a result, there is neither an increment left to capitalize in land values nor a reduction in ground values caused by the tax.

By contrast, we may see what occurs if the number of towns with schools is strictly limited. In figure 2, the assumption is made that only two towns have junior high schools, when it would require three towns to hold all of the people who would wish to use them, assuming prices were set equal to the average educational cost within those two towns. In this case, the effective supply curve is $mnsS'$, and the market-clearing price is r. Assuming the towns simply try to cover school costs with their taxes rather than making a profit for the public fisc, average taxes will be m, and a consumer's surplus of *mrlsn* would be available for those lucky enough to reside in those towns. However, the oversupply of would-be residents at price m would be expected to drive the price up to r. In this case, the area *mnsr* would represent the portion of consumer benefits captured by land rents. There would be, here, net capitalization of a portion of school benefits above real average costs. On the other hand, if a third town were then to open a school—at the same cost per pupil m—land value capitalization would disappear. The eventual situation would be as in figure 1, with the prior capitalization a temporary phenomenon due to the quicker approach to equilibrium by the first towns to open schools. Thus, if there are long-run constant returns to scale in service provision, an equilibrium can be approached as in the Tiebout model, but as that equilibrium is reached differential land values will be removed.

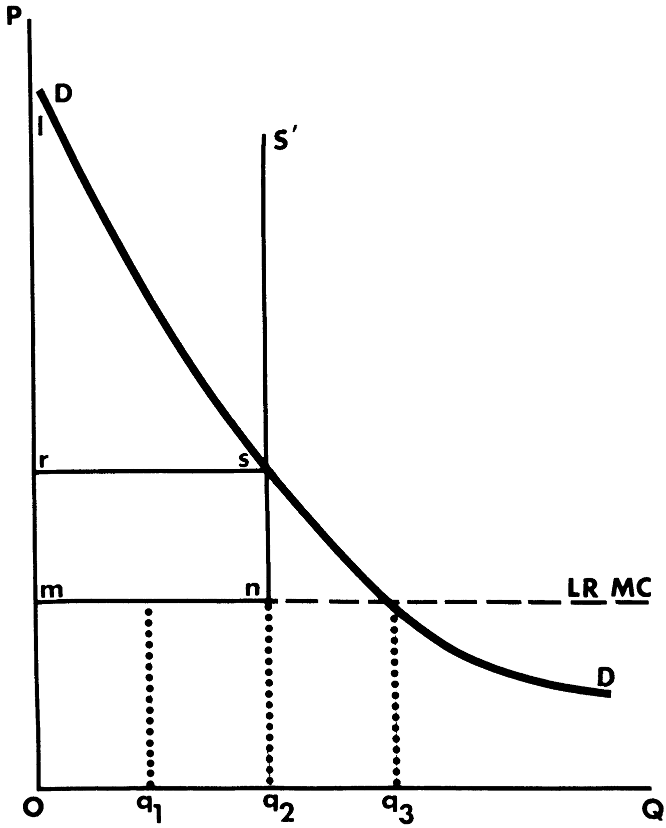

Fig. 2.—A market for schools with supply restricted to two towns

Relaxation of the assumption of constant returns alters the result only slightly. In figure 3, average and marginal costs rise with the number of users or the amount of the service provided. Such a situation may arise if production of the service uses scarce resources other than land (as in the case of health service provision if the supply of doctors and nurses is not perfectly elastic). If the service is sold at its marginal cost, as in a private goods market, the market will insure optimal resource allocation. There will be a producer's surplus *lrs*. This producer's surplus will take the form of land values, if one has to be a landowner to supply the service, as Grieson (1971) shows is the case for services of buildings. If, however, the service is paid for through taxation, hence sold on an average cost basis, the quantity purchased would be q_e, and while consumer's surplus would be greater producer's surplus would be eliminated. Nor would any of the consumer's surplus be capitalized into land values, for the prospective consumers would all be in equilibrium, paying only a price (the tax) equal to marginal utility (marginal demand).

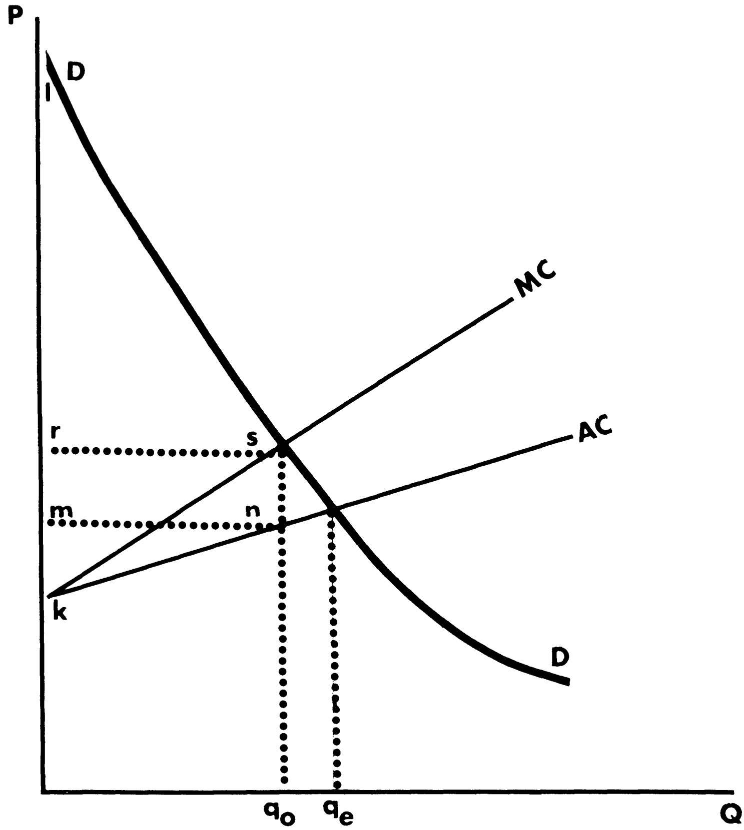

FIG. 3.—A market for medical facilities with doctors a scarce good

Capitalization will occur if some nonprice mechanism is used to limit demand to q_o, the optimal quantity at which demand intersects marginal cost. If, for example, zoning is used to limit entry into the school district, but taxes continue to cover average costs alone, then, for those consumers able to enter, marginal benefits will be at the level r while tax costs will equal m. The difference will be left as a pure consumer surplus (*mnrs* in addition to the original *lrs*) to families allowed in, if rationing (e.g., racial discrimination) is used to limit entry. However, if land is sold on a market, subject only to a density-zoning constraint, then the price per user must rise to r, and the area *mnrs* is left capitalized into land values. (This area is equal to the producer's surplus *krs* in the case of market sale of the service itself.)

In short, if only one service at a time is considered, the first five assumptions of the Tiebout model lead to a situation in which, at equilibrium, there is no capitalization effect. Taxes cover costs simply, and there is neither a net increment in land values from the presence of the service nor a decrease from the presence of the tax. Since the tax is a form of average-cost pricing of the service, the equilibrium may or may not be Pareto optimal. In the case of constant returns to scale, presence of capitalization indicates inefficiency. If there are decreasing returns to scale, the presence of positive capitalization is compatible with efficiency (although it does not guarantee that a situation of efficiency exists).[2] In the case of increasing returns to scale, by a similar argument, the provision of efficient service levels financed out of average-cost local taxation would involve negative capitalization.[3] But in these last two cases, efficiency (if it were to exist) would require methods of limiting or attracting entrants to the community through methods not foreseen in the Tiebout model. Zoning might, in this case, make a situation either more or less optimal. To show that there is capitalization in a model with only one public service thus fails to show that a Tiebout mechanism has assured either equilibrium or efficiency.

The situation is more complex if different public services are present within a system of competing towns (see Buchanan and Goetz 1972). In this case, if some services are in equilibrium and some not in equilibrium, different total levels of taxation may be associated with different levels of land values, and presence of services whose provision and financing are in equilibrium may appear to be associated with higher land values. Suppose one service (let us say schooling) is provided by a Tiebout system that is in equilibrium, but there are other services which are not provided in equilibrium. The difference between marginal demand and average cost for these other services will be capitalized. If, however, a comparison of towns, using regression analysis, treats the total tax rate as a single variable, then the net noncapitalization of school taxes and expenditures may appear as offsetting positive capitalization of the expenditures and negative capitalization of the taxes. That is, by comparing the school taxes with other taxes that are unmatched by benefits, the two elements of capitalization are conceptually separable. But if average-cost taxes and marginal benefits are equal in the provision of all services, there will be no way of making this separation by comparison of the towns. If the Tiebout model held perfectly (at least in its first five assumptions) for all services, a regression analysis relating land values to taxes and service levels, holding other background conditions constant, would show no capitalization of any taxes or services.

[2] The Pareto optimal situation in this case may involve a more unequal distribution of benefits than a case of "inefficient" oversupply of the service where nobody is excluded by nonmarket means. The problems involved in applying the Tiebout model to cases of income redistributing public services are explored by Rothenberg (1969, 1970) and by Miller and Tabb (1973).

[3] One parallel to this is Goldberg's (1970, 1972) demonstration that a transportation system under certain assumptions is most efficient if it minimizes land values (see also Edel 1971).

III

The preceding discussion has indicated that a world in which the conditions of the Tiebout model were met perfectly would see no tax and local expenditure capitalization into property values. Oates sought to demonstrate the "relevance to actual behavior" of the Tiebout model by a demonstration of capitalization using New Jersey data for 1960. His linking of an observable quantity, property value, to the Tiebout model is a fruitful step in public finance theory. However, the version of the Tiebout model which Oates tests is oversimplified.

Tiebout, it will be remembered, presented his theory as a response to the Musgrave-Samuelson agreement that "no 'market type' solution exists to determine the level of expenditures on public goods. Seemingly we are faced with the problem of having a rather large portion of our national income allocated in a 'non-optimal' way when compared with the private sector" (Tiebout 1956). Indeed, his very title was a play on Samuelson's 1954 article. Tiebout's paper sought to show that, at least conceptually (under what he admitted were "extreme assumptions"), a "market type" solution might allow the optimal provision of public services out of local taxes.

Oates, on the other hand, simplifies the Tiebout model into a statement about consumer demand only, with no supply side made explicit, and no optimality issue raised: "Very simply, Tiebout's world is one in which the consumer 'shops' among different communities offering varying packages of local public services and selects as a residence the community which offers the tax-expenditure program best suited to his tastes" (Oates 1969, p. 957). His results, therefore, are never compared with the results predicted from a full Tiebout model. They are, rather, a demonstration that taxes and services have some effect on consumer preferences for town of residence, which is reflected in house prices. The tax and service capitalization which Oates finds would not occur if consumers were completely indifferent to services and taxes, had no information about their differences among towns, or succeeded in passing all local taxes on completely to the federal government in reduced income taxes. Oates eliminates these possibilities, demonstrating the individual economic rationality of the home buyer.[4] But his results also would not occur if the pure theory as presented by Tiebout were a complete description for all local service markets.

In the specific case of school expenditures, Oates calculates, using "typical" values, that the amount of decrease in home values from capitalization of approximately two-thirds of a 1 percent tax rate increase would roughly equal the increase in home prices from the spending of the

[4] Other studies have used analyses of price responsiveness to demonstrate that peasants are "rational" (for discussion see Edel 1970).

money raised on an average school. The calculation is presented "as indicating no more than orders of magnitude" and as a suggestion that, tax and expenditure capitalization being equal, the market for schooling may be in Tiebout's equilibrium. However, if the interest rate used for capitalization or the two-thirds rate of capitalization were different, the lack of net capitalization of schooling would not hold. A Tiebout equilibrium for education is thus demonstrated only on the basis of strong assumptions, while that at least one sector of town expenditures must be out of equilibrium is conclusively demonstrated by the finding of any capitalization. A similar result follows from the similar finding of capitalization by Heinberg and Oates (1970) using 1960 Massachusetts data.

Oates concludes from his analysis: "These results appear consistent with a model of the Tiebout variety in which rational consumers weigh (to some extent at least) the benefits from local public services against the cost of their tax liability in choosing a community of residence; people do appear willing to pay more to live in a community which provides a high-quality of public services (or in a community which provides the same program of public services with lower tax rates)" (Oates 1969, p. 968). This conclusion is warranted, but it is incomplete. "A model of the Tiebout variety" as used by Oates does not consider the elasticity or inelasticity of supply and hence whether the welfare-economic implications of Tiebout's original model hold. By introducing the analysis of capitalization into the discussion of Tiebout's model, Oates has taken a significant step. However, it is necessary to go beyond his single-period measure of capitalization to examine changes in capitalization during supply adjustment in order to determine whether a Tiebout equilibrium is being approached. In the following section, Oates's capitalization approach is used to examine such a historical process.

IV

Examination of data on house values and public finance for a group of towns in the Boston metropolitan area over several decades allows such an examination of the supply-adjustment capitalization process.[5] Boston area data provide an appropriate test because of the extreme dependence of towns in that vicinity on the local property tax for financing services. In Massachusetts, 54 percent of local expenditures were financed from this tax as of 1970, as compared to a 40 percent national average. Highway and educational maintenance expenditures, the two funding categories to be considered in detail, have particularly low levels of outside aid. The

[5] This study is part of a longer historical examination of factors influencing Boston-area property values.

figure for highway expenditures, it should be noted, does not include construction of new roads or highways, for which outside funding is considerable, but only their upkeep, which is a considerable expense in a snow-affected area like New England.

Using a somewhat different sample from the Massachusetts data than Heinberg and Oates, we estimated regressions of a somewhat similar form to theirs, for the 5 census years beginning with 1930. These regressions use six distance variables (latitudinal distance from downtown Boston, longitudinal distance, and their squares and cubes) rather than a single accessibility variable. A variable for per-square-mile highway maintenance expenditures is a second form of public expenditure, along with school expenditures per pupil, whose capitalization into house values is tested. Density and percentage of owner occupiers among residents of a town are used to capture amenity factors not explained by distance. The home-ownership term may also capture the shifting of the property tax through federal income tax deductions. Tax rates are nominal rates for 1930–60, and both nominal and equalized rates are employed in separate regressions for 1970.

Estimated coefficients appear in table 1 and are superficially similar to Oates's findings for the postwar period. Tax-rate capitalization is negative in all years and significant for 1940, 1950, and 1970. School expenditures per pupil are capitalized positively in all 3 postwar census years. However, for 1960 and 1970, the effect is not statistically significant at the .01 level and is quantitatively smaller than for 1950. For 1930 and 1940 the school-expenditure term is not significant, and indeed the sign is negative. The highway-maintenance expenditures are not significantly capitalized in any period.

The regressions appear to show somewhat weaker capitalization than Oates's examples. Assuming a \$21,000 house with a tax rate of \$50 per \$1,000 (a typical figure 5 miles north or south of the city in the 1970 regressions), a \$10 reduction in the effective tax rate would amount to a \$210 reduction in taxes. Capitalized at an 8 percent interest rate, this would amount to a \$2,625 increase in value, about twice the observed increase given by the regression estimate. In other words, tax capitalization in this case would be about 50 percent, rather than the two-thirds found by Oates. Perhaps more important than this difference is the declining payoff in house values from each dollar per pupil spent on schools: the capitalization falls from almost \$24 in 1950 to under \$9 in 1960 and about \$3 in 1970. At first glance, this might appear to show that Bostonians are less concerned with education and taxes than are Oates's New Jersey residents and that they are becoming less concerned with schools over time.

An alternative explanation suggests an approach, over time, to a Tiebout equilibrium in the provision of schooling. According to Hirsch

TABLE 1

REGRESSION ESTIMATES OF EFFECT OF TAXES, SCHOOL AND ROAD EXPENDITURES, AND OTHER VARIABLES ON HOUSE PRICES

Year	Tax Rate ($/$1,000)	Density (Pop./Sq. Mile)	Owner Occupied	School Expenditures ($/Pupil)	Highway Maintenance ($/Sq. Mile)	R^2 (n)
1930	−503 (1.31)	0.050 (0.73)	n.a.	−2.31 (0.28)	−0.006 (0.44)	.349 (78)
1940	−651 (2.49)	−0.015 (0.33)	−5.01 (0.40)	−3.55 (0.67)	−5.01 (0.40)	.355 (77)
1950	−811 (3.03)	−0.017 (0.17)	52.48 (1.99)	23.68 (4.08)	−0.009 (0.49)	.648 (77)
1960	−126 (0.54)	−0.400 (2.56)	0.001 (0.03)	8.77 (1.59)	0.008 (0.54)	.376 (73)
1970	−43.3 (2.18)	−0.374 (1.49)	86.67 (1.41)	2.99 (1.48)	0.012 (0.74)	.546 (65)
1970* ...	−139.0* (2.06)	−0.211 (0.59)	127.87 (2.10)	3.25 (1.60)	0.023 (1.39)	.551 (64)

NOTE.—These are values of b from regressions in which latitudinal and longitudinal distance and their squares and cubes were also independent variables. House value is measured in dollars from census estimates of town medians. (Values of t are in parentheses.)
* Using equalized tax rates (other regressions use nominal tax rates).

(1968), schooling is a service which can be provided efficiently by relatively small units. Economies of scale exist only up to sizes of perhaps a few hundred pupils for elementary schools and 1,700 for secondary schools. These sizes are within the population range of most Boston area communities; the few that are too small to support high schools can easily form consolidated high school districts. There are also sufficient independent towns in each geographic region of the Standard Metropolitan Statistical Area to allow consumer choice between towns with expensive schools and towns with low school expenditures which attract childless individuals or parents preferring private and parochial schools. (Among the higher-income communities, the comparison between Brookline and Belmont is most obvious.) The conditions cited by Tiebout as assumptions in his model might thus seem to hold for schools. This explanation is consistent with the 1970 regressions: assuming each family, on the average, has three children of school age, $1.00 per pupil spent by the schools would represent a $3.00 benefit to the family (if taxes and other community and housing characteristics are held constant). The $3.00 increase in house values is thus an approximation to a simple capitalization of the benefits, which would be matched exactly by an offsetting loss of $3.00 *if* families paid average cost through taxes and if taxes were fully capitalized. In practice, the equilibrium is not exactly obtained: tax capitalization appears only partial, the average (although perhaps not the marginally moving) family has less than three children, and some school expenditures are borne by state and federal aid rather than local property taxes. Nonetheless, comparing the 3 years—1950, 1960, and 1970—it is the low capitalization figure for school expenditures in 1970 that is approximately consistent with a Tiebout equilibrium.

But what, then, of the higher capitalization in benefits for 1950 and 1960? It would appear that these represent higher returns due to a shortage of schools. A situation obtains such as that in figure 2 rather than figure 1; due to limited supply the marginal demand for schooling is far above the average cost. Beginning in the early postwar period, that is to say, there appears to have been a shortage of adequate quantities and qualities of schools, given the baby boom, the rising need for technical skills in the labor force, and the much-noted shortage of teachers and school facilities. In this situation, parents would pay far more than the actual average cost of schooling through increased house prices in order to enter a community with what they considered to be adequate educational facilities. By 1960, the shortage was somewhat reduced from the 1950 level, and capitalization of per-pupil expenditures had declined somewhat. By 1970, there appears to have been very little markup, if any, left. The comparison of the three periods thus seems to be much like the movement from a short-run to a long-run equilibrium in Marshallian economics.[6]

Tiebout's model thus seems to have been a fair representation of the schooling market in the Boston area by 1970. Oates, measuring returns in 1960 both for New Jersey and (with Heinberg) for Massachusetts, would seem to have found greater returns because he took a cross-section view before equilibrium was achieved. His speculation that "it could be . . . the negative association we have observed between property taxes and home values is primarily a short-run phenomenon, which would disappear over a longer period of time" (Oates 1969, p. 967) has thus been borne out. Indeed the logic of the Tiebout model indicates this should occur in an approach to equilibrium.

The figures in table 1 also indicate, however, that while the provision of schooling may have approached a Tiebout equilibrium, the same could not have been true for all other local services. In 1970 tax rates (whether viewed in real or nominal terms) significantly affected house values, even though they were not 100 percent capitalized. What is more, local road-maintenance expenditures were not positively capitalized to a statistically significant extent in any period, although some positive effect is found for 1970. If all expenditures were in a Tiebout equilibrium, taxes should not have been capitalized, while, since taxes were capitalized, road maintenance should have had positive effects on house values if it had positive benefits from a consumer viewpoint. Thus not all markets

[6] The explanation for the negative capitalization of school expenditures for 1930 and 1940 may be that, because of the depression, some communities were left with school average costs above marginal benefits, which they were unable to reduce.

can be in Tiebout equilibria, and the road-maintenance market appears likely as a candidate to be out of equilibrium.

Road expenditures are as clear an example of violation of Tiebout's conditions as schools are a conforming example. Neither are there clear constant returns to scale, nor is there an absence of spillovers. Indeed, in the Boston area, highway-maintenance expenditures appear mainly related to the flow-through of traffic and population density. Highway expenses in dollars per square mile (HXP) for 1964 are explained by the regression:

$$\text{HXP} = 34857 + \underset{(10.01)}{31.62}\ \text{FLOW} + \underset{(10.94)}{11.27}\ \text{RES}\ (R^2 = .784),$$

where FLOW = commuter flow-through per square-mile area, and RES = residual population density estimated from:

$$\text{DENSITY} = 2567 + \underset{(10.29)}{4.00}\ \text{FLOW}\ (R^2 = .631).$$

Addition of a term for median per-family income has no significant effect. Inner-ring communities must maintain roads through which out-of-town commuters pass although their residents get few benefits. Only a few road expenses, such as frequency of snow removal in winter on side streets, are really optional expenses that may rise with income or local preference. Thus, differences in road-maintenance expenditures do not represent a response to demand differences, as in the Tiebout model, and are not capitalized.[7] The same is true for some other road-related services such as traffic-police expenditure and contributions to the Metropolitan District Commission highway system.

Thus, despite the movement toward a Tiebout-type equilibrium for schooling, there appear to be public service markets (including road maintenance, for one) which do not show movement toward such an equilibrium. The Tiebout hypothesis appears to hold for some public service markets. But, since it does not hold for all of them, prudence (reinforced by the general theory of the second best) would caution against acceptance of the notion that a "market-type" mechanism of the Tiebout sort has beneficial efficiency properties if applied to all public services taken together, as compared with any possible alternative systems of public-expenditure determination for metropolitan areas.

References

Buchanan, J. M., and Goetz, C. J. "Efficiency Limits of Fiscal Mobility: An Assessment of the Tiebout Model." *J. Public Econ.* 1 (April 1972): 25–43.

Edel, M. "Innovative Supply: A Weak Point in Economic Development Theory." *Soc. Sci. Information* 9 (June 1970): 9–40.

[7] Further discussion of commuter-related expenses is found in Sclar (1971).

Edel, M. "Land Values and the Costs of Urban Congestion: Measurement and Distribution." *Soc. Sci. Information* 10 (December 1971): 7–36.

Goldberg, M. A. "Transportation, Urban Land Values, and Rents: A Synthesis." *Land Econ.* 46 (May 1970): 153–62.

———. "An Evaluation of the Interaction between Urban Transport and Land Use Systems." *Land Econ.* 48 (November 1972): 338–46.

Grieson, R. "The Economics of Property Taxes and Land Values." Massachusetts Inst. Tech., Department of Economics Working Paper, 1971.

Heinberg, J. D., and Oates, W. E. "The Incidence of Differential Property Taxes on Urban Housing." *Nat. Tax J.* 23 (March 1970): 92–98.

Hirsch, W. Z. "The Supply of Urban Public Services." In *Issues in Urban Economics*, edited by H. S. Perloff and L. Wingo, Jr. Baltimore: Johns Hopkins Univ. Press, 1968.

Miller, S., and Tabb, W. K. "A New Look at a Pure Theory of Local Expenditures." *Nat. Tax J.* 26 (June 1973): 161–76.

Musgrave, R. A. *The Theory of Public Finance.* New York: McGraw-Hill, 1959.

Oates, W. E. "The Effects of Property Taxes and Local Spending on Property Values: An Empirical Study of Tax Capitalization and the Tiebout Hypothesis." *J.P.E.* 77 (November/December 1969): 957–71.

Rothenberg, J. "Strategic Interaction and Resource Allocation in Metropolitan Intergovernmental Relations." *A.E.R.* 59 (May 1969): 495–503.

———. "Local Decentralization and the Theory of Optimal Government." In *The Analysis of Public Output*, edited by J. Margolis. New York: Columbia Univ. Press (for Nat. Bur. Econ. Res.), 1970.

Samuelson, P. A. "The Pure Theory of Public Expenditures." *Rev. Econ. and Statis.* 36 (November 1954): 387–89.

Sclar, E. "The Public Financing of Transportation in the Greater Boston Area." Ph.D. dissertation, Tufts Univ., 1971.

Tiebout, C. M. "A Pure Theory of Local Expenditures." *J.P.E.* 65 (October 1956): 416–24.